

Solutions to Dummit & Foote

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1 Section 0

Section 0.1

Exercise 1.1. $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathcal{B} = \{X \in \mathcal{A} : MX = XM\}$

$$\begin{aligned} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} &= \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

By the calculations above, only $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are in $\mathcal{B}$.

Note. Multiplying M to the left amounts to the row operation that adds the second column to the first column. Multiplying M to the right amounts to the column operation that adds the first column to the second column..

Exercise 1.2. Suppose $P, Q \in \mathcal{B}$. Then

$$\begin{aligned} M(P + Q) &= MP + MQ && \text{(distribution law)} \\ &= PM + QM && \text{(assumption)} \\ &= (P + Q)M. && \text{(distribution law)} \end{aligned}$$

Therefore, $P + Q \in \mathcal{B}$.

Exercise 1.3. Suppose $P, Q \in \mathcal{B}$. Then

$$\begin{aligned} M(PQ) &= (MP)Q && \text{(associative law)} \\ &= (PM)Q && \text{(assumption)} \\ &= P(MQ) && \text{(associative law)} \\ &= P(QM) && \text{(assumption)} \\ &= (PQ)M. && \text{(associative law)} \end{aligned}$$

Therefore, $PQ \in \mathcal{B}$.

Exercise 1.4. By brute-force calculation,

$$\begin{aligned} M \begin{pmatrix} p & q \\ r & s \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p+r & q+s \\ r & s \end{pmatrix} \\ \begin{pmatrix} p & q \\ r & s \end{pmatrix} M &= \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p+q \\ r & r+s \end{pmatrix} \end{aligned}$$

So, we solve

$$\begin{pmatrix} p+r & q+s \\ r & s \end{pmatrix} = \begin{pmatrix} p & p+q \\ r & r+s \end{pmatrix}.$$

Solving the set of equations yields $r = 0, p = s$.

Exercise 1.5. (a) $f(a/b) = a$ is not well-defined, e.g. $1/1 = 2/2$, while $f(1/1) = 1 \neq 2 = f(2/2)$. Therefore, equivalent fractions may be assigned different values by f .

(b) $f(a/b) = a^2/b^2$ is well-defined. For any fractions $a/b = c/d$, we have $a^2/b^2 = c^2/d^2$ by squaring both sides. Thus $f(a/b) = a^2/b^2 = c^2/d^2 = f(c/d)$. Hence f is well-defined.

Exercise 1.6. The function f is not well-defined. Consider the following two decimal expansion of $r = 1 \in \mathbb{R}^+$.

$$1.000\ldots \quad \text{and} \quad 0.999\ldots$$

It is a standard fact of real analysis that both are decimal expansions of 1. Applying the definition of f to the two decimal expansions above of r will yield 0 and 9, respectively. Hence f is not well-defined.

Exercise 1.7. The binary relation $\sim$ on A is defined by

$$a \sim b \text{ if and only if } f(a) = f(b)$$

For any $a \in A$, the equivalence class of a is

$$[a]_{\sim} = \{x \in A : a \sim x\} = \{x \in A : f(a) = f(x)\}$$

By definition, $[a]_{\sim}$ is the fiber of f over $f(a)$. Since any equivalence class arise in this way, every equivalence class of $\sim$ is a fiber of f . Conversely, let $b \in B$. By the surjectivity of f , there exists an element $a \in A$ such that $f(a) = b$. Then $[a]_{\sim}$ is the fiber of f over $f(a) = b$, as it has shown above. Therefore, the fiber $f^{-1}(b)$ is the equivalence class $[a]_{\sim}$. Since every fiber of f arise in this way, every fiber of f is an equivalence class of $\sim$. This completes the proof.

Section 0.2

1. For each of the following pairs of integers a and b , determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y .

- (a) $a = 20, b = 13$.
- (b) $a = 69, b = 372$.
- (c) $a = 792, b = 275$.
- (d) $a = 11391, b = 5673$.
- (e) $a = 1761, b = 1567$.
- (f) $a = 507885, b = 60808$.

Exercise 2.1. We'll use the formula $dl = ab$ to calculate the least common divisor.

(a) $a = 20, b = 13$. By euclidean algorithm,

$$\begin{aligned} 20 &= 1 \cdot 13 + 7 \\ 13 &= 1 \cdot 7 + 6 \\ 7 &= 1 \cdot 6 + 1 \\ 6 &= 6 \cdot 1 + 0 \end{aligned} \tag{1}$$

The gcd of 20 and 13 is equal to 1

The lcm of 20 and 13 is $20 \cdot 13/1 = 260$. By equation 1 $1 = 2 \cdot 20 - 3 \cdot 13$.

(b)

$$\begin{aligned}69 &= 0 \cdot 372 + 69 \\372 &= 5 \cdot 69 + 27 \\69 &= 2 \cdot 27 + 15 \\27 &= 1 \cdot 15 + 12 \\15 &= 1 \cdot 12 + 3 \\12 &= 4 \cdot 3 + 0\end{aligned}\tag{2}$$

The gcd of 69 and 372 is equal to 3

The lcm of 69 and 372 is $69 \cdot 372/3 = 8556$. From equation 2, $27 \cdot 69 - 5 \cdot 372 = 3$.

(c) $a = 792$, $b = 275$. By the Euclidean algorithm,

$$\begin{aligned}792 &= 2 \cdot 275 + 242 \\275 &= 1 \cdot 242 + 33 \\242 &= 7 \cdot 33 + 11 \\33 &= 3 \cdot 11 + 0\end{aligned}\tag{3}$$

The gcd of 792 and 275 is equal to 11

The lcm of 792 and 275 is $792 \cdot 275/11 = 19800$. By Equation (3), $11 = 8 \cdot 792 - 23 \cdot 275$.

(d) $a = 11391$, $b = 5673$. By the Euclidean algorithm,

$$\begin{aligned}11391 &= 2 \cdot 5673 + 45 \\5673 &= 126 \cdot 45 + 3 \\45 &= 15 \cdot 3 + 0\end{aligned}\tag{4}$$

The gcd of 11391 and 5673 is equal to 3

The lcm of 11391 and 5673 is $11391 \cdot 5673/3 = 21540381$. By Equation (4), $3 = -126 \cdot 11391 + 253 \cdot 5673$.

(e) $a = 1761$, $b = 1567$. By the Euclidean algorithm,

$$\begin{aligned}1761 &= 1 \cdot 1567 + 194 \\1567 &= 8 \cdot 194 + 15 \\194 &= 12 \cdot 15 + 14 \\15 &= 1 \cdot 14 + 1 \\14 &= 14 \cdot 1 + 0\end{aligned}\tag{5}$$

The gcd of 1761 and 1567 is equal to 1

The lcm of 1761 and 1567 is $1761 \cdot 1567 = 2759487$. By Equation (5), $1 = -105 \cdot 1761 + 118 \cdot 1567$.

(f) $a = 507885$, $b = 60808$. By the Euclidean algorithm,

$$\begin{aligned}507885 &= 8 \cdot 60808 + 21421 \\60808 &= 2 \cdot 21421 + 17966 \\21421 &= 1 \cdot 17966 + 3455 \\17966 &= 5 \cdot 3455 + 691 \\3455 &= 5 \cdot 691 + 0\end{aligned}\tag{6}$$

The gcd of 507885 and 60808 is equal to 691

The lcm of 507885 and 60808 is $507885 \cdot 60808 / 691 = 44693880$. By Equation (6), $691 = -17 \cdot 507885 + 142 \cdot 60808$.

Exercise 2.2. Suppose k divides a and b , so that $a = ka_0$ and $b = kb_0$ for $a_0, b_0 \in \mathbb{Z}$. Then for any $s, t \in \mathbb{Z}$, $as + bt = k(a_0s + b_0t)$. Therefore, k divides $ds + bt$.

Exercise 2.3. Suppose n is composite. Then there exists a divisor of n greater than 1, say a . Then $b = n/a$ is also an integer. Then we certainly n divides $ab = n$, but n does not divide a nor b since $a, b < n$.

Exercise 2.4. For any integer t ,

$$\begin{aligned} ax + by &= a(x_0 + \frac{b}{d}t) + b(y_0 - \frac{a}{d}t) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N. \end{aligned}$$

Therefore,

$$x = x_0 + \frac{b}{d}t \text{ and } y = y_0 - \frac{a}{d}t \tag{7}$$

are also solutions to $ax + by = N$.

To show that equation 7 is the general solution, suppose $ax + by = N$. As $ax_0 + by_0 = N$, $a(x - x_0) + b(y - y_0) = 0$. Thus $\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0)$. We need to show that $\frac{b}{d}$ divides $x - x_0$. Since $\frac{b}{d}$ divides the RHS, it divides $\frac{a}{d}(x - x_0)$. Since $\frac{b}{d}$ and $\frac{a}{d}$ are relatively prime, $\frac{b}{d}$ divides $x - x_0$. So write $x - x_0 = \frac{b}{d}t$ for some $t \in \mathbb{Z}_0$. Then $y - y_0 = -\frac{a}{d}t$.

Exercise 2.5.

$$\begin{aligned}\phi(1) &= 1 \\ \phi(2) &= 2 - 1 = 1 \\ \phi(3) &= 3 - 1 = 2 \\ \phi(4) &= \phi(2^2) = 2(2 - 1) = 2 \\ \phi(5) &= 5 - 1 = 4 \\ \phi(6) &= \phi(2)\phi(3) = 2 \\ \phi(7) &= 7 - 1 = 6 \\ \phi(8) &= \phi(2^3) = 2^2(2 - 1) = 4 \\ \phi(9) &= \phi(3^2) = 3(3 - 1) = 6 \\ \phi(10) &= \phi(2)\phi(5) = 4 \\ \phi(11) &= 11 - 1 = 10 \\ \phi(12) &= \phi(3)\phi(2^2) = 2 \cdot 2 = 4 \\ \phi(13) &= 13 - 1 = 12 \\ \phi(14) &= \phi(2)\phi(7) = 6 \\ \phi(15) &= \phi(3)\phi(5) = 8 \\ \phi(16) &= \phi(2^4) = 2^3(2 - 1) = 8 \\ \phi(17) &= 17 - 1 = 16 \\ \phi(18) &= \phi(2)\phi(3^2) = 6 \\ \phi(19) &= 19 - 1 = 18 \\ \phi(20) &= \phi(2^2)\phi(5) = 2 \cdot 4 = 8 \\ \phi(21) &= \phi(3)\phi(7) = 12 \\ \phi(22) &= \phi(2)\phi(11) = 10 \\ \phi(23) &= 23 - 1 = 22 \\ \phi(24) &= \phi(2^3)\phi(3) = 4 \cdot 2 = 8 \\ \phi(25) &= \phi(5^2) = 5(5 - 1) = 20 \\ \phi(26) &= \phi(2)\phi(13) = 12 \\ \phi(27) &= \phi(3^3) = 3^2(3 - 1) = 18 \\ \phi(28) &= \phi(2^2)\phi(7) = 2 \cdot 6 = 12 \\ \phi(29) &= 29 - 1 = 28 \\ \phi(30) &= \phi(2)\phi(3)\phi(5) = 1 \cdot 2 \cdot 4 = 8.\end{aligned}$$

Exercise 2.6. Let $A \subset \mathbb{Z}^+$ be a nonempty subset. We prove by induction that for any $n \in \mathbb{Z}^+$, if $n \in A$, then A has a minimal element.

For $n = 1$, if $1 \in A$, then 1 is clearly the minimal element of A .

Assume the statement holds for $n = k$. Suppose A contains an element at most $k + 1$.

If $k + 1 \notin A$, then A contains an element at most k , so by the induction hypothesis A has a minimal element.

If $k + 1 \in A$, there are two possibilities. If A contains no element less than $k + 1$, then $k + 1$ is the minimal element of A . Otherwise, A contains an element at most k , and by the induction hypothesis there is a minimal element m among those elements. Since $m < k + 1$, m is also the minimal element of A .

Thus the statement holds for $k + 1$, so it holds for all $n \in \mathbb{Z}^+$.

Now let A be a nonempty subset of $\mathbb{Z}^+$ and choose some $a \in A$. Applying the result to $n = a$, we conclude that A has a minimal element.

To show that a minimal element in A is unique, suppose both m and m' are minimal elements of A . Then $m \leq m' \leq m$ by the definition of minimal elements. Therefore, $m = m'$.

2 Section 1

Subsection 1

Exercise 1.1.